The Last Useful Feature

Propensity Neighborhoods for Energy Program Choice

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1 Setting

Suppose a utility can estimate every eligible program's enrollment propensity for every customer. The note can be summarized in two sentences:

Adding feature coordinates to a customer representation can only shrink its fixed-radius neighborhood. Useful refinement therefore requires coordinates that preserve propensity similarity and enough remaining peers for the gain in match quality to exceed sampling uncertainty.

This simple fact connects representation, local evidence, and recommendation. Behavioral history can create coordinates that describe customers well. A propensity model turns those coordinates into one comparable score row per customer. Nested neighborhoods then show how much detail the observed evidence can sustain, while a finite-sample bound supplies a stopping rule and a confidence test for the final program choice.

The experiments make this division of labor visible. Behavioral embeddings substantially improve future propensity estimates and program rankings, showing that the choice of coordinates matters most. Neighborhoods then sharpen local calibration and reveal when further refinement has too little support to justify changing the ranking. Practically, this theory-to-method package provides a disciplined way to learn customer coordinates, retrieve well-supported peers, and decide when local evidence is strong enough to alter or qualify a program recommendation.

2 From behavior to a propensity matrix

Let $x_1, \dots, x_N$ be N customers and let $\mathcal{A} = \{a_1, \dots, a_K\}$ be a common catalog of K programs. Thus $i \in \{1, \dots, N\}$ indexes customers and $j \in \{1, \dots, K\}$ indexes programs. Eligibility is applied before an individual recommendation, giving the feasible subset $\mathcal{A}(x) \subseteq \mathcal{A}$ for target customer x. For any customer x and program a, define the true enrollment propensity

$$p_a(x) = \Pr(x \text{ enrolls in program } a) \in [0, 1].$$

Collect the K catalog propensities into the row vector $p(x) = (p_{a_j}(x) : j = 1, \dots, K) \in [0, 1]^K$. The true customer-by-program matrix is $P = (p_{a_j}(x_i)) \in [0, 1]^{N \times K}$. A fitted model supplies an estimate $\widetilde{p}_a(x)$ of each probability and hence the fitted matrix

$$\widetilde{P} = \begin{bmatrix} \widetilde{p}_{a_1}(x_1) & \widetilde{p}_{a_2}(x_1) & \cdots & \widetilde{p}_{a_K}(x_1) \\ \widetilde{p}_{a_1}(x_2) & \widetilde{p}_{a_2}(x_2) & \cdots & \widetilde{p}_{a_K}(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \widetilde{p}_{a_1}(x_N) & \widetilde{p}_{a_2}(x_N) & \cdots & \widetilde{p}_{a_K}(x_N) \end{bmatrix} \in [0, 1]^{N \times K}.$$

Thus the row $\widetilde{p}(x) = (\widetilde{p}_{a_j}(x) : j = 1, \dots, K) \in [0, 1]^K$ is a fitted propensity vector. A tilde marks a fitted quantity throughout. The base recommender masks ineligible columns and ranks the remaining entries:

$$\widetilde{a}(x) \in \arg\max_{a \in \mathcal{A}(x)} \widetilde{p}_a(x). \tag{1}$$

2.1 A behavioral embedding

Customer attributes need not be the only coordinates. Let $H = (H_{i\ell}) \in \mathbb{R}^{N \times J}$ record pre-decision behavior over J items or events, where $H_{i\ell}$ is customer i's bounded engagement with item or event $\ell \in \{1, \dots, J\}$. Choose an embedding dimension $D \le \min\{N, J\}$. A rank-D factorization,

$$H \approx U_D \Sigma_D V_D^\top, \qquad z(x_i) = (U_D \Sigma_D)_{i\cdot} \in \mathbb{R}^D,$$

where $U_D \in \mathbb{R}^{N\times D}$ and $V_D \in \mathbb{R}^{J\times D}$ contain customer and item directions, $\Sigma_D \in \mathbb{R}^{D\times D}$ contains their singular values, $^\top$ denotes transpose, and $(\cdot)_{i\cdot}$ selects row i. This maps a long behavioral history to a D-coordinate customer embedding $z(x_i)$ [2]. The factorization is learned from history only. If $s(x)$ denotes the vector of ordinary premise and customer attributes, a program model f_a takes the form

$$\widetilde{p}_a(x) = f_a\big(s(x), z(x)\big), \qquad a \in \mathcal{A}. \tag{2}$$

Any supervised or unsupervised embedding may replace the factorization; what matters below is whether distance in its coordinates preserves future propensity similarity.

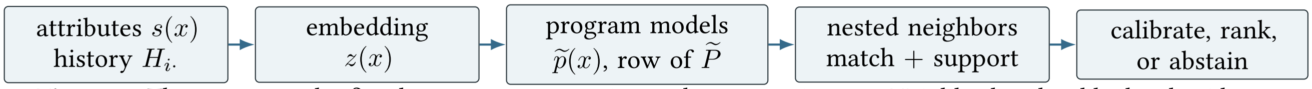

Figure 1. The recommender first learns a representation and a propensity row. Neighborhoods add a local evidence layer around that row.

3 Nested neighborhoods

Let x now denote a target customer and let $x_1, \dots, x_n$ denote n historical reference customers, renumbered for convenience from the larger customer population. Choose a positive integer M and order the available representation into M blocks. Blocks may contain static attributes, groups of embedding coordinates, or groups of fitted propensity coordinates. Let $d_m(x, x') \ge 0$ be the cumulative distance after the first m blocks and require

$$d_{m+1}(x, x') \ge d_m(x, x'), \qquad m = 1, \dots, M-1.$$

Equivalently, set $\delta_1(x, x') = d_1(x, x')$ and $\delta_j(x, x') = d_j(x, x') - d_{j-1}(x, x') \ge 0$ for $j \ge 2$, so that $d_m(x, x') = \sum_{j=1}^m \delta_j(x, x')$. This includes cumulative Euclidean, block-additive, and maximum-coordinate distances. For a fixed radius $r \ge 0$, set

$$\mathcal{N}_m(x; r) = \{i \in \{1, \dots, n\} : d_m(x, x_i) \le r\}, \qquad n_m = |\mathcal{N}_m(x; r)|.$$

Only nonempty levels are considered.

Proposition 1 (More detail leaves fewer peers)**.** *For $m = 1, \dots, M-1$,*

$$\mathcal{N}_{m+1}(x; r) \subseteq \mathcal{N}_m(x; r), \qquad n_{m+1} \le n_m.$$

Proof. Because $\delta_{m+1} \ge 0$, $d_{m+1}(x, x') \ge d_m(x, x')$. A customer within radius r after block $m+1$ was therefore within radius r after block m. □

The refinement cost is exact: each added block can spend peer support. To describe its benefit, let $g(x) = (g_a(x) : a \in \mathcal{A}) \in \mathbb{R}^K$ be any catalog-indexed target quantity whose local homogeneity matters and define

$$\varepsilon_m^g(x) = \max_{i \in \mathcal{N}_m(x; r)} \|g(x_i) - g(x)\|_\infty,$$

where $\|v\|_\infty = \max_{a\in\mathcal{A}} |v_a|$; for a target-specific certificate, the maximum may be restricted to $a \in \mathcal{A}(x)$. For direct peer averaging take $g = p$. For residual calibration take

$$g(x) = \rho(x) := p(x) - \widetilde{p}(x),$$

the vector of errors remaining after the base propensity model; its program-a coordinate is $\rho_a(x) = p_a(x) - \widetilde{p}_a(x)$.

Proposition 2 (Refinement improves worst-case local similarity)**.** *If $\mathcal{N}_{m+1}(x; r)$ is nonempty, then*

$$\varepsilon_{m+1}^g(x) \le \varepsilon_m^g(x).$$

Proof. By Proposition 1, the later neighborhood is a subset of the earlier one. The maximum of the same nonnegative function over a subset cannot increase. □

The fitted propensity matrix gives a particularly compact geometry. Full-row matching uses

$$d_p(x, x_i) = \|\widetilde{p}(x) - \widetilde{p}(x_i)\|_\infty.$$

When calibrating a particular program a, a leave-one-program-out alternative uses

$$d_{-a}(x, x_i) = \max_{b \in \mathcal{A} \setminus \{a\}} |\widetilde{p}_b(x) - \widetilde{p}_b(x_i)|.$$

Both are available once $\widetilde{P}$ has been scored. Progressive groups of propensity coordinates also satisfy Proposition 1.

4 The refinement frontier

For reference customer $i \in \{1, \ldots, n\}$ and program $a \in \mathcal{A}(x)$, let $Y_{ia} \in [0, B]$, where $B > 0$, be a bounded calibration outcome satisfying $\mathbb{E}[Y_{ia} \mid x_i] = p_a(x_i)$. For binary enrollment, $B = 1$. Historical fitted scores $\widetilde{p}_a(x_i)$ are computed out of fold or on a separate fitting sample. At level m, use the peers' mean residual to correct the target score:

$$\widehat{p}_{a,m}(x) = \widetilde{p}_a(x) + \frac{1}{n_m} \sum_{i \in \mathcal{N}_m(x;r)} \big(Y_{ia} - \widetilde{p}_a(x_i)\big), \qquad \widehat{a}_m \in \arg\max_{a \in \mathcal{A}(x)} \widehat{p}_{a,m}(x). \tag{3}$$

The estimate may be clipped to $[0, 1]$ coordinatewise. Let $a^* \in \arg\max_{a \in \mathcal{A}(x)} p_a(x)$ be a true best eligible program, let $K_x = |\mathcal{A}(x)|$ be the number of eligible programs, and write $\varepsilon^\rho_m(x)$ for the residual mismatch defined above.

Theorem 3 (Residual refinement frontier)**.** *Suppose that, conditional on the customer profiles, the calibration outcomes $\{Y_{ia} : i = 1, \ldots, n\}$ are independent for each a, and that the neighborhoods are fixed independently of these outcomes. For a prespecified level m with $n_m \geq 1$ and confidence parameter $\alpha \in (0, 1)$, with probability at least $1 - \alpha$,*

$$p_{a^*}(x) - p_{\widehat{a}_m}(x) \leq 2b_m(x), \qquad b_m(x) = \varepsilon^\rho_m(x) + B\sqrt{\frac{\log(2K_x/\alpha)}{2n_m}}.$$

If one of M levels is selected using the same calibration outcomes, replace K_x by $K_x M$.

Proof. Let $\bar{\rho}_{a,m}(x) = n_m^{-1} \sum_{i \in \mathcal{N}_m(x;r)} \rho_a(x_i)$ be the peers' mean true residual. Taking conditional expectations in (3) gives $\widetilde{p}_a(x) + \bar{\rho}_{a,m}(x)$. Since $p_a(x) = \widetilde{p}_a(x) + \rho_a(x)$,

$$|\bar{\rho}_{a,m}(x) - \rho_a(x)| \leq \varepsilon^\rho_m(x).$$

Hoeffding's inequality and a union bound over the K_x eligible programs bound the deviation of each empirical mean residual from $\bar{\rho}_{a,m}(x)$ by $q_m = B\sqrt{\log(2K_x/\alpha)/(2n_m)}$. Thus $|\widehat{p}_{a,m}(x) - p_a(x)| \leq b_m(x)$ simultaneously in a. Because $\widehat{a}_m$ maximizes the fitted values,

$$p_{a^*}(x) - p_{\widehat{a}_m}(x) \leq [p_{a^*}(x) - \widehat{p}_{a^*,m}(x)] + [\widehat{p}_{\widehat{a}_m,m}(x) - p_{\widehat{a}_m}(x)] \leq 2b_m(x).$$

A second union bound over M levels proves the adaptive statement. □

The two terms move in opposite directions. Proposition 2 makes the residual mismatch nonincreasing; Proposition 1 makes the uncertainty term nondecreasing. Let $\widehat{\varepsilon}^\rho_m(x) \geq 0$ denote an estimate of $\varepsilon^\rho_m(x)$ obtained from separate validation data or cross-fitted residuals. A literal finite-sample certificate uses an upper confidence estimate; a validation estimate gives the corresponding tuning rule. The customer-specific choice is

$$\widehat{m}(x) \in \arg\min_{1 \leq m \leq M} \left\{ \widehat{\varepsilon}^\rho_m(x) + B\sqrt{\frac{\log(2K_x M/\alpha)}{2n_m}} \right\}. \tag{4}$$

Corollary 4 (A confident recommendation)**.** *Assume $K_x \geq 2$. On the event in Theorem 3, let $\widehat{a}_m$ have empirical lead*

$$\Delta_m(x) = \widehat{p}_{\widehat{a}_m,m}(x) - \max_{a \in \mathcal{A}(x) \setminus \{\widehat{a}_m\}} \widehat{p}_{a,m}(x).$$

If $\Delta_m(x) > 2b_m(x)$, then $\widehat{a}_m$ is the unique true best program.

Proof. Every true propensity lies within b_m of its estimate. An empirical lead exceeding $2b_m$ therefore remains positive after both scores move by their largest permitted errors. □

Shrinkage is often useful in implementation. Multiplying the residual correction in (3) by $\lambda \in [0,1]$ changes the uniform error term to at most

$$(1-\lambda)\|\rho(x)\|_\infty + \lambda \varepsilon_m^\rho(x) + \lambda B \sqrt{\frac{\log(2K_x/\alpha)}{2n_m}}.$$

Cross-validation can select λ before the test period. Corollary 4 then turns the bound into a practical gate: rerank when local evidence clears the top-two margin, otherwise retain the base choice or mark it low confidence.

5 A recommender workflow

The mathematics fits into a six-step operating loop. Each step has an operational object and, where relevant, a formal guarantee.

1. **Generate candidates and filter eligibility.** Construct the feasible program set before ranking:

$$\mathcal{A}(x) = \{a : a \text{ is eligible, permitted, and currently available to } x\}.$$

   Eligibility, contact, capacity, and timing rules determine $\mathcal{A}(x)$; propensity ranking begins after that set is formed.

2. **Represent the customer.** Build the behavioral embedding from information available before the recommendation and combine it with ordinary attributes:

$$H \approx U_D \Sigma_D V_D^\top, \qquad z(x_i) = (U_D \Sigma_D)_{i\cdot}, \qquad v(x) = \big(s(x), z(x)\big).$$

   The representation supplies the geometry. Its coordinates are useful when closeness in $v(x)$ preserves future propensity or residual similarity.

3. **Score and rank the propensity row.** Evaluate every eligible program and collect the fitted probabilities:

$$\widetilde{p}_a(x) = f_a(v(x)), \qquad \widetilde{p}(x) = \big(\widetilde{p}_a(x) : a \in \mathcal{A}(x)\big), \qquad \widetilde{a}(x) \in \arg\max_{a \in \mathcal{A}(x)} \widetilde{p}_a(x).$$

   Evaluate the same models out of fold on historical customers. Their rows in $\widetilde{P}$ are directly comparable with the target row and can support local calibration without in-sample residuals.

4. **Retrieve increasingly detailed neighborhoods.** Add declared blocks of attributes, embedding coordinates, or propensity coordinates:

$$d_m(x, x_i) = \sum_{j=1}^{m} \delta_j(x, x_i), \qquad \mathcal{N}_1(x;r) \supseteq \mathcal{N}_2(x;r) \supseteq \cdots \supseteq \mathcal{N}_M(x;r).$$

   Proposition 1 fits here: adding nonnegative coordinate contributions can only remove fixed-radius peers. The next stage searches only the IDs that survived the previous stage.

5. **Measure match, support, and useful depth.** At every level record

$$n_m = |\mathcal{N}_m(x;r)|, \qquad \widehat{\varepsilon}_m^\rho(x), \qquad \widehat{m}(x) \in \arg\min_{1 \le m \le M} \left\{ \widehat{\varepsilon}_m^\rho(x) + B\sqrt{\frac{\log(2K_x M/\alpha)}{2n_m}} \right\}.$$

   Proposition 2 and Theorem 3 fit here: refinement improves the worst remaining match while lost peers increase uncertainty. The selected level is the last point at which the trade remains favorable.

6. **Calibrate, recommend, or retain the base ranking.** Use the selected peers' out-of-fold residuals:

$$\widehat{p}_{a,\widehat{m}}(x) = \widetilde{p}_a(x) + \frac{1}{n_{\widehat{m}}} \sum_{i \in \mathcal{N}_{\widehat{m}}(x;r)} \big(Y_{ia} - \widetilde{p}_a(x_i)\big), \qquad \widehat{a}(x) \in \arg\max_{a \in \mathcal{A}(x)} \widehat{p}_{a,\widehat{m}}(x).$$

Compute the top-two margin $\Delta_{\widehat{m}}(x)$. If it exceeds $2b_{\widehat{m}}(x)$, issue the locally supported recommendation; otherwise retain the direct model choice or attach a low-confidence flag. *Corollary 4 fits here: the margin converts the error bound into a decision certificate.*

The computation is naturally staged. Cheap coordinates screen the reference population; later blocks are evaluated only for surviving peers and only for customers whose leading programs remain close. The theorem therefore governs statistical effort and computational effort through the same shrinking set.

6 Chronological evidence from KuaiRec

The framework was tested on KuaiRec, a nearly fully observed short-video recommendation dataset with 1,411 users and 3,327 items [1]. Video categories play the role of programs, and a customer's propensity for category a is the fraction of its videos watched to completion. The twelve largest categories form the action set and supply a behavioral analogue of program enrollment.

Each user's interactions were ordered by time. The first 60% formed the observable history H; the last 40% formed future propensity targets. Within each of ten repeated outer splits, 60% of users estimated ridge propensity models and out-of-fold residuals, 25% tuned neighborhood choices, and 15% were untouched test users. A 32-coordinate truncated-SVD embedding was fit using estimation histories only. Approximately 2.22 million early interactions built the representations and 1.48 million later interactions supplied outcomes. Operational comparisons used $k \in \{25, 50, 100, 200\}$ nearest neighbors, where k is the fixed peer count, and validation-selected residual shrinkage. This fixes peer support and complements the fixed-radius refinement studied by the theorem. RMSE averages coordinatewise error against the observed future category rates; top-1 regret is the future-rate difference between the best category and the recommended category; hit rate is the fraction for which those categories coincide.

Table 1. Mean chronological test performance over ten splits. Lower RMSE and regret are better; higher hit rate is better. Panel B uses ten fresh splits after its full-row rule was frozen.

Method	RMSE	Top-1 regret	Hit rate
Panel A: behavioral representation			
Static customer features	0.126761	0.041357	34.46%
Static features + embedding	0.086310	0.038726	37.04%
Embedding-space shared neighbors	0.085960	**0.038530**	**37.09%**
Embedding-space program-specific neighbors	**0.085564**	0.039276	36.48%
Panel B: propensity-space confirmation			
Static features + embedding	0.083850	**0.036119**	**39.30%**
Full-row propensity neighbors	**0.083498**	0.036931	38.83%
Leave-one-program-out neighbors	0.083833	0.037950	37.51%

The behavioral representation supplies the main gain. Relative to static customer features, adding the embedding reduces propensity RMSE by 31.9%, reduces top-1 regret by 6.4%, and raises the exact hit rate by 2.58 percentage points. RMSE improves in all ten splits, regret in eight, and hit rate in nine. This confirms the paper's central modeling premise: coordinates matter because they determine which customers can be meaningfully close.

The neighborhood experiments separate calibration from ranking. Embedding-space neighbors reduce regret by 6.8% relative to static features across all ten splits, although their incremental gain over the embedding model is small. In the fresh propensity-space confirmation, full-row neighbors improve RMSE in nine of ten splits. They also show that better calibration need not change the winning program favorably: their mean regret is 2.2% above the direct embedding ranker. Leave-one-program-out matching is less informative, which indicates that a program's own score is an important coordinate of propensity-space retrieval.

An earlier static-feature pilot displayed the geometric mechanism directly. Deeper fixed-radius neighborhoods lost coverage and performance, while the support-aware rule selected the first three levels for 71.1%, 17.7%, and 11.2% of customers and never selected the deepest level. Its certificate covered realized regret, though it was numerically loose. Taken together, the experiments give the theorem three concrete roles: reject unhelpful coordinate systems, stop refinements before support disappears, and gate score corrections by the recommendation margin.

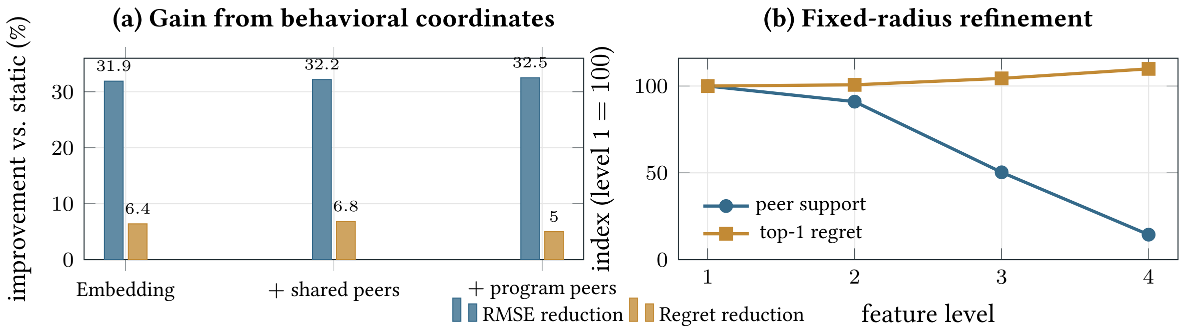

Figure 2. Experimental geometry. Panel (a) reports percentage improvements over the static propensity model in the ten-split chronological experiment; positive values are better. Panel (b) indexes the first pilot's fixed-radius support and regret to level 1. Added feature blocks remove peers rapidly, and recommendation regret rises once refinement outruns support.

7 Remarks on the result

The geometry is modest and useful: it identifies the price of asking for a closer match. Behavioral embeddings improve the propensity model because their coordinates preserve future preference. Propensity rows then provide a compact twelve-coordinate surface for cohort retrieval and local calibration. The margin certificate determines when that local evidence is strong enough to affect the recommendation.

Enrollment propensity and causal value answer different questions. The present result concerns enrollment propensity. If the objective is incremental enrollment or energy savings, $p_a(x)$ can be replaced by an identified action-specific treatment effect under the corresponding assignment and policy-evaluation design [4]. Customer targeting can materially affect program savings and cost effectiveness [3].

The propositions say exactly what refinement does. The theorem prices what it costs. The embedding and propensity matrix show where the method fits in a working recommender. The conclusion is the two-sentence summary with which the note began:

Adding feature coordinates to a customer representation can only shrink its fixed-radius neighborhood. Useful refinement therefore requires coordinates that preserve propensity similarity and enough remaining peers for the gain in match quality to exceed sampling uncertainty.

References

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